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Snowflake Agentic ELT Solution for Financial Analytics

Executive Overview

The Agentic ELT application is an AI based ***no-code*** solution for developing data pipelines, it empowers finance and analytics teams to deploy sophisticated **ELT pipelines and workflows** using the familiar tech stack of snowflake and dbt (in-house) without the burden of manual data engineering. This solution fundamentally transforms fragmented financial tables into a unified data foundation, topping it with an a modular, agent-based approach to:

1. **Enrich** Bronze data with metadata (L0).
2. **Generate** standardized SQL for Silver views (L1).
3. **Generate** incremental dbt SQL for materialized tables (L2).
4. **Empower** users with a conversational agent to build and deploy validated Gold tables (L3) as per business requirements.

The agentic ELT pipeline uses **self-healing**, SQL validation and **Human-In-The-Loop** concepts to retain reliability and accountability in the process.

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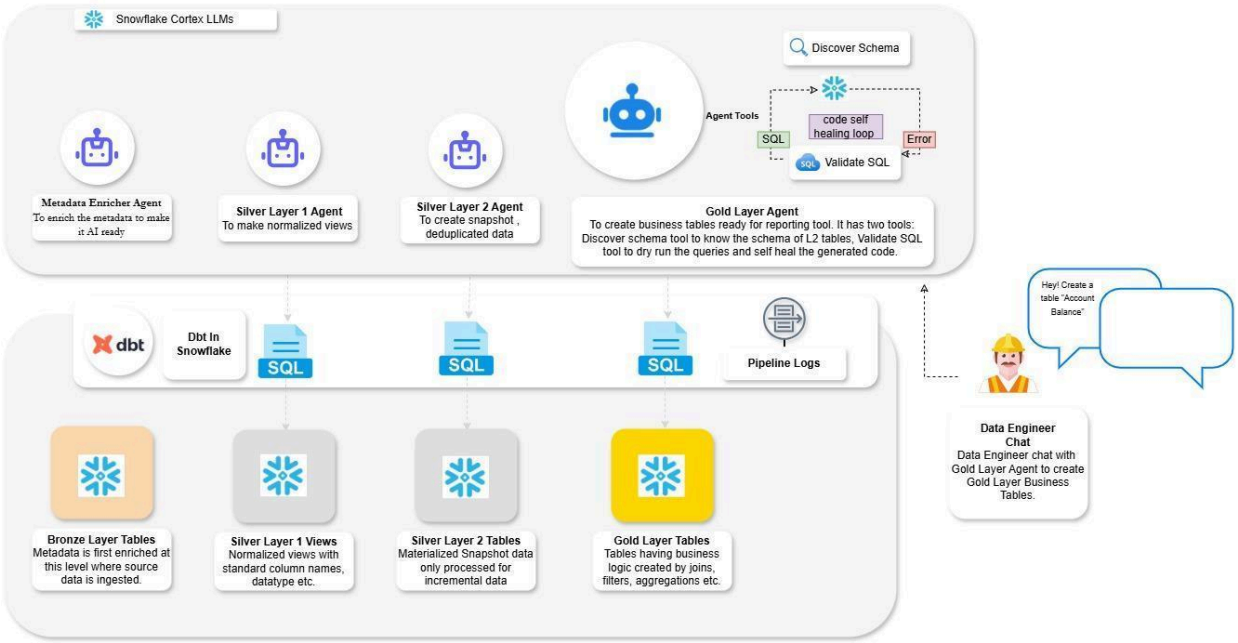
Agentic ELT Pipeline : Architecture Overview



The entire pipeline follows the standard Medallion Architecture (Bronze -> Silver -> Gold), augmented by intelligent agents at each step.

The flow is as follows:

Bronze (L0): Raw data is ingested.

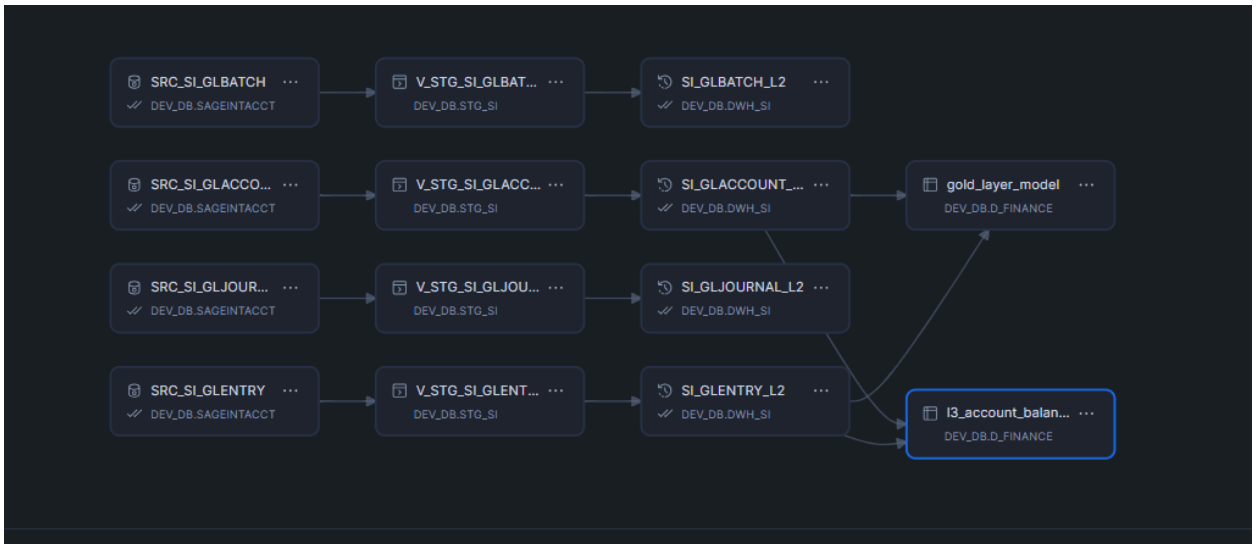
Metadata Enricher: An agent scans Bronze tables and adds table/column comments using a human-in-the-loop (HITL) process.

Silver (L1): An agent generates SQL to create standardized views (e.g., proper column names) which are executed using dbt (in-house).

Silver (L2): An agent assists a human in defining incremental logic and generates dbt models.

Gold (L3): An advanced conversational agent (with robust validation and self-healing) chats with an engineer to build business-ready tables, also outputting dbt SQL model.

Dbt in Snowflake : The execution of the models, quality checks and lineage mapping is handled by in-house dbt.



Construction & Data Strategy used for the application testing:

This solution is built on the Snowflake Data Cloud, providing an intelligent yet reliable solution. Using the four foundational pillars—**GLACCOUNT, GLENTRY, GLBATCH,** and **GLJOURNAL**—it establishes a strong foundation for agent-driven finance analytics workflows.

Data Model Summary

Table	Description	Records
SI_GLENTRY_L2	GL transaction entries with amounts, dates, locations	245,909
SI_GLACCOUNT_L2	Chart of accounts master data	450
SI_GLBATCH_L2	Journal batch headers	6,021
SI_GLJOURNAL_L2	Journal definitions	25

Key Features

Agent-Driven Automation (Reduced Manual Effort)

- Automates end-to-end ELT workflows including table onboarding, SQL generation, and pipeline orchestration
- Minimizes manual coding through metadata-driven and agent-assisted model creation
- Accelerates development cycles and reduces dependency on most of the real-time engineering tasks

Built-in Validations & Data Trust

- Incorporates multi-layer validations across Bronze, Silver, and Gold to ensure data quality
- Schema checks, null handling, and transformation-level validations improve reliability
- Ensures consistent and trustworthy outputs for downstream analytics and reporting

Human-in-the-Loop Governance

- Critical steps (e.g., SQL generation, table approvals) include user validation checkpoints
- Enables controlled automation with the ability to review, approve, or override decisions
- Balances speed with governance, ensuring production-grade confidence

Reliable & Resilient Pipelines

- Designed for idempotent and repeatable builds to avoid data duplication or inconsistencies
- Handles incremental processing efficiently for large-scale datasets
- Integrates seamlessly with CI/CD (Github) workflows for consistent deployments

Lineage-Aware Transformations

- Tracks dependencies across models and layers for full data lineage visibility
- Enables traceability from source (Bronze) to business-ready outputs (Gold)
- Simplifies impact analysis, debugging, and auditability

Scalable & Domain-Agnostic Framework

- Easily extensible to multiple domains such as Finance (Sage Intacct), HR (Workday), or ERP (Oracle)
- Modular design supports rapid onboarding of new data sources and use cases
- Built to scale with growing data volumes and complexity

Business Value

- ⏱️ ***Faster Time-to-Insight***
- 💸 ***Reduced Engineering Costs***
- 📈 ***Improved Data Reliability & Trust***
- 🔍 ***Enhanced Transparency & Governance***
- 🚀 ***Scalable Across Domains***
- 🔄 ***Increased Operational Efficiency***
- 🤝 ***Business–Tech Alignment***

The Dataplatr Advantage

The Dataplatr **Agentic ELT Solution for Financial Analytics** provides an agentic way of deploying your data pipelines. By combining Agentic ELT automation with Dataplatr's deep ***expertise in financial customization***, we help enterprises modernize ERP workflows and achieve faster time-to-insight.

Contact us at: **info@dataplatr.com**

For consultations or custom inquiries: https://dataplatr.com/contact-us

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